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A Project Report on

"Operating System"

Submitted in partial fulfillment for the requirements for the first semester Bsc.CSIT in Introduction of Science and technology

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UNDER THE GUIDANCE OF

(Lecturer)

(Introduction of Science and technology, SNSC)

SUBMITTED TO

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Introduction

Operating system (OS)

An operating system (OS) is the program that, after being initially loaded into the computer by a boot program, manages all of the other application programs in a computer. The application programs make use of the operating system by making requests for services through a defined application program interface (API). In addition, users can interact directly with the operating system through a user interface, such as a command-line interface (CLI) or a graphical UI (GUI).

Why use an operating system?

An operating system is essential because it acts as the vital bridge between complex computer hardware and the everyday user. Without it, you would have to write manual code in binary just to perform basic tasks like moving a mouse or saving a file. It serves as a resource manager, efficiently distributing CPU power and memory so you can multitask without system crashes. Additionally, it provides a user-friendly interface (GUI) and a secure environment, protecting your data through encryption and user permissions while ensuring that various software applications can communicate seamlessly with different hardware devices via drivers.

OBJECTIVE

The main obejectives are given below:

- To Define Core Concepts: Understanding the fundamental role of an OS as an interface between the hardware and the user.
- To Understand Resource Management: Demonstrate how the system manages the CPU (Processors), RAM (Memory), and Storage (File Systems) to prevent crashes.
- To Analyze OS Architecture: Detail the relationship between the Kernel (the heart), the Shell (the interface), and Hardware.
- To Compare OS Types: Evaluate different types of systems, such as Batch, Real-Time, and Mobile OS, to show how they serve different needs.
- To Identify Pros and Cons: Discuss the advantages (like multitasking and ease of use) and disadvantages (like cost and security risks) of modern systems.
- To Explore Performance Metrics: Understand how an OS improves throughput (work done) and minimizes response time for the user.

METHODS OF STUDY

I collected primary as well as secondary data by using various methods. I followed the following methods to collect the primary data about recent Operating System.

- Literature Review: Studying standard and technical manuals to understand OS theory.
- Comparative Analysis: Comparing different systems like Windows, Linux, and macOS to identify differences in their kernels and file systems.
- System Observation: Using tools like Task Manager (Windows) or Top (Linux) to monitor real-time CPU and RAM usage.
- Virtualization: Using Virtual Machines (like VirtualBox or VMware) to install and test different operating systems safely on one computer.
- Documentation Research: Reviewing official developer documentation (like Microsoft Docs or Linux Kernel Archives) for technical details.

FINDINGS

By collecting various data, we found following things:

- History Of OS
- Types of Operating System (OS)
- Functions of Operating System
- Features of Operating System (OS)
- Advantage of using Operating System • Disadvantages of using Operating System
- What is a Kernel?
- Difference between Firmware and Operating System
- Difference between 32-Bit vs. 64 Bit Operating System

History Of OS

The history of operating systems is typically divided into generations that mirror the evolution of computer hardware, from room-sized vacuum tube machines to modern smartphones and AI-driven systems.

1. First Generation (1945–1955): Vacuum Tubes & No OS

Technology: These computers used vacuum tubes and plugboards for programming.

Operation: There was no operating system. Programs were manually entered in absolute machine language (binary) by physically moving switches or using plugboards.

Use: Mostly simple numerical calculations for scientific purposes.

2. Second Generation (1955–1965): Transistors & Batch Systems

Technology: Transistors replaced vacuum tubes, making machines more reliable and smaller.

Key Milestone: The first recognized operating system, GM-NAA I/O, was introduced in 1956 for the IBM 704.

Batch Processing: Similar jobs were collected on punch cards and processed as a "batch" to reduce CPU idle time.

3. Third Generation (1965–1980): Integrated Circuits & Multitasking

Technology: Integrated Circuits (ICs) allowed for much faster processing.

Multiprogramming: OS designers developed the ability to keep several jobs in memory at once; if one job waited for I/O, the CPU switched to another.

Unix Revolution: In 1969, Unix was developed at Bell Labs, introducing a modular design that influenced almost all modern operating systems, including Linux and macOS.

Time-Sharing: Systems like Multics allowed hundreds of users to access a single large computer simultaneously.

4. Fourth Generation (1980–Present): Personal Computers & GUIs

Technology: Large-scale integration (LSI) put thousands of transistors on a single chip, leading to the microcomputer.

MS-DOS (1981): Microsoft launched MS-DOS for the IBM PC, which dominated the early personal computer market.

GUI Revolution: Apple's Macintosh (1984) popularized the Graphical User Interface (icons and mouse), which was later followed by Microsoft Windows (1985).

Linux (1991): Linus Torvalds released the open-source Linux kernel, which now powers most web servers and Android devices.

5. Fifth Generation (2000s–Present): Mobile & AI Integration

Mobile OS: The rise of smartphones led to iOS (2007) and Android (2008), optimized for touchscreens and battery efficiency.

Cloud & AI: Modern systems like Windows 11 and ChromeOS are heavily integrated with cloud services and AI assistants like Siri, Alexa, and Google Assistant

Types of Operating System (OS)

Following are the popular types of Operating System:

- Batch Operating System
- Multitasking/Time Sharing OS
- Multiprocessing OS
- Real Time OS
- Distributed OS
- Network OS
- Mobile OS

Batch Operating System

Some computer processes are very lengthy and time-consuming. To speed the same process, a job with a similar type of needs batched together and run as a group.

The user of a batch operating system never directly interacts with the computer. In this type of OS, every user prepares his or her job on an offline device like a punch card and submit it to the computer operator.

Multi-user

These systems allow two or more users to run programs at the same time. Some OS permit hundreds or even thousands of concurrent users. The operating systems of mainframes and minicomputer are multi-user systems. Examples are MVS, UNIX, etc. Another term for multi-user is time-sharing.

Multiprocessing

It refers to a computer system‟s ability to support more than one process (program) at the same time. This system allows the simultaneous execution of programs on a computer that has several CPUs. Mainframe, Supercomputers have more than one CPU.

Multitasking

This system allows a single CPU to execute what appears to be more than one program at the same time when, in fact, only one program is being executed. In multitasking, only one CPU is involved, but it switches from one program to another so quickly that it gives the appearance of executing all of the programs at the same time.

Multithreading

Multithreading allows different parts of a single program to run concurrently. Multithreading is the ability of an OS to execute different parts of a program, called threads, simultaneously.

Distributed Operating System

Distributed systems use many processors located in different machines to provide very fast computation to its users.

Network Operating System

Network Operating System runs on a server. It provides the capability to serve to manage data, user, groups, security, application, and other networking functions.

Mobile OS

Mobile operating systems are those OS which is especially that are designed to power smartphones, tablets, and wearables devices.

Some most famous mobile operating systems are Android and iOS, but others include BlackBerry, Web, and watchOS.

Functions of Operating System

Below are the main functions of Operating System:

In an operating system software performs each of the function:

1. **Process management: -** Process management helps OS to create and delete processes. It also provides mechanisms for synchronization and communication among processes.
2. **Memory management: -** Memory management module performs the task of allocation and de-allocation of memory space to programs in need of this resources.
3. **File management: -** It manages all the file-related activities such as organization storage, retrieval, naming, sharing, and protection of files.
4. **Device Management:** Device management keeps tracks of all devices. This module also responsible for this task is known as the I/O controller. It also performs the task of allocation and de-allocation of the devices.

5. **I/O System Management:** One of the main objects of any OS is to hide the peculiarities of that hardware devices from the user.
6. **Secondary-Storage Management:** Systems have several levels of storage which includes primary storage, secondary storage, and cache storage. Instructions and data must be stored in primary storage or cache so that a running program can reference it.
7. **Security: -** Security module protects the data and information of a computer system against malware threat and authorized access.
8. **Command interpretation:** This module is interpreting commands given by the and acting system resources to process those commands.
9. **Networking:** A distributed system is a group of processors which do not share memory, hardware devices, or a clock. The processors communicate with one another through the network.
10. **Job accounting:** Keeping track of time & resource used by various job and users.
11. **Communication management:** Coordination and assignment of compilers, interpreters, and another software resource of the various users of the computer systems.

Features of Operating System (OS)

Here is a list important features of OS:

- Protected and supervisor mode
- Allows disk access and file systems Device drivers Networking Security
- Program Execution
- Memory management Virtual Memory Multitasking
- Handling I/O operations
- Manipulation of the file system
- Error Detection and handling
- Resource allocation
- Information and Resource Protection

Advantage of using Operating System

1. User-Friendly Interface (GUI)

- **No Coding Needed:** Users don’t need to know programming or binary code to use a computer. You can just click icons, drag files, and use menus.
- **Consistency:** Most apps look and act similarly (like having a "File" menu in both Word and Photoshop) because the OS provides a standard template.

2. Resource Sharing & Management

- **Multitasking:** You can listen to music, browse the web, and write a report at the same time. The OS manages the CPU so all these apps feel like they are running at once.
- **Hardware Abstraction:** You don't have to write different code for every printer or monitor. The OS uses **drivers** to make all hardware work "out of the box."

3. Data Security & Protection

- **Access Control:** The OS uses usernames and passwords to keep your private files safe from other users.
- **Firewalls & Defender:** Modern OSs come with built-in tools to block viruses and unauthorized hackers.

4. Memory & Storage Organization

- **Efficient RAM Usage:** The OS ensures that if one app crashes, it doesn't take down the entire system by "walling off" that app's memory.
- **Indexing:** The File System allows you to find a specific document among millions of files in seconds.

5. Automatic Updates & Fixes

- **Self-Healing:** Modern operating systems can detect errors, fix corrupted files, and download security patches automatically to keep the system stable.

Disadvantages of using Operating System

1. High System Requirements

- **Resource Heavy:** Modern OSs like Windows 11 require a lot of **RAM**, **storage**, and **CPU** power just to run the background tasks, leaving less for your actual apps.
- **Obsolescence:** As OSs get updated, older hardware becomes "obsolete" and too slow to handle the new features.

2. Vulnerability to Attacks

- **Single Point of Failure:** Since the OS controls everything, if a **virus** or **malware** gets into the Kernel, it can take over your entire computer, steal data, or spy on you.
- **Security Patches:** Users must constantly download updates to fix "bugs" or security holes found by hackers.

3. Cost Factor

- **Expensive Licenses:** While Linux is free, major operating systems like **Windows** or **macOS** (built into the price of a Mac) can be very expensive for businesses and individuals.
- **Subscription Models:** Some features now require monthly fees or "Pro" versions.

4. System Crashes (The "Blue Screen")

- **Stability Issues:** If a core part of the OS (like a driver) fails, the entire system can crash, causing you to lose unsaved work.
- **Complexity:** Because they are made of millions of lines of code, they are never 100% bug-free.

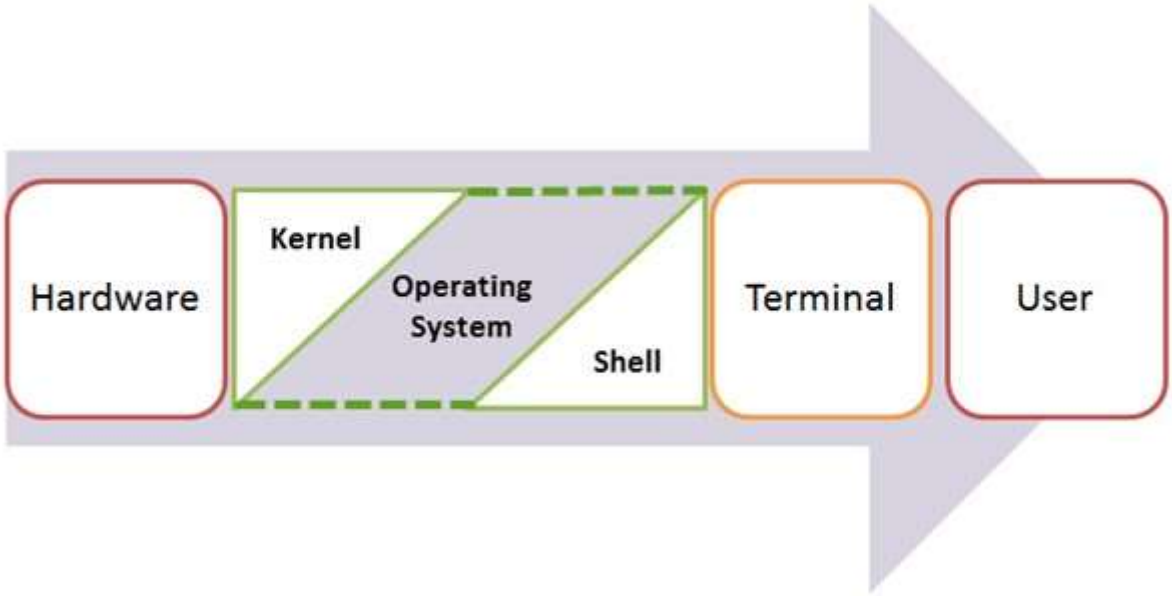
5. Privacy Concerns

- **Data Tracking:** Many modern OSs (especially mobile ones like Android) track user behavior, location, and app usage to show targeted ads or improve services, which some users find intrusive.

Kernel

The kernel is the central component of a computer operating systems. The only job performed by the kernel is to the manage the communication between the software and the hardware. A Kernel is at the nucleus of a computer. It makes the communication between the hardware and software possible.

While the Kernel is the innermost part of an operating system, a shell is the outermost one.



The Shell (The Interpreter)

The Shell is the layer that wraps around the kernel. It acts as an interpreter that takes your commands and translates them into a language the kernel understands.

CLI (Command Line Interface): Typing commands like in Terminal or Command Prompt.

GUI (Graphical User Interface): The visual layer with icons, windows, and menus that you click.

System Calls (The Bridge)

System Calls are the "request forms" that programs use to ask the kernel for help.

Example: When you click "Save" in Word, the app sends a System Call to the kernel asking for permission to write data to the hard drive.

File System (The Librarian)

The File System defines how data is labeled, stored, and retrieved.

What it does: It keeps track of where a file "lives" on the disk so the OS doesn't have to search the entire drive every time you open a photo.

Device Drivers (The Translators)

Drivers are specific bits of code that tell the OS how to communicate with external hardware.

Analogy: If the OS is a conductor, the Drivers are the sheet music that tells him how to play a specific "instrument" (like a printer or graphics card).

Difference between 32-Bit vs. 64 Bit Operating System

Parameters	32. Bit	64. Bit
Architecture and Software	Allow 32 bit of data processing simultaneously	Allow 64 bit of data processing simultaneously
Compatibility	32-bit applications require 32-bit OS and CPUs.	64-bit applications require a 64-bit OS and CPU.
Systems Available	All versions of Windows 8, Windows 7, Windows Vista, and Windows XP, Linux, etc.	Windows-XP Professional, Vista, 7, Mac OS X and Linux.
Memory Limits	32-bit systems are limited to 3.2 GB of RAM.	64-bit systems allow a maximum 17 billion GB of RAM.

CONCLUSION

In conclusion, the operating system serves as the indispensable backbone of modern computing, transforming raw hardware into a functional and intuitive tool. Through its core functions of process management, memory allocation, and security, it ensures that a computer operates with maximum efficiency and stability. While different types of systems like Windows, Linux, and Android cater to various user needs, they all share the fundamental goal of providing a seamless interface that hides technical complexity. As technology continues to evolve with AI and cloud integration, the role of the operating system remains critical in managing resources and protecting data, proving that it is truly the "soul" of the machine.

REFERENCE

- Microsoft Documentation. "How Operating Systems Work." learn.microsoft.com.
- The Linux Foundation. "Inside the Linux Kernel." kernel.org.
- https://hamrocsit.com/
- https://www.geeksforgeeks.org/operating-systems/operating-systems/